SenStar

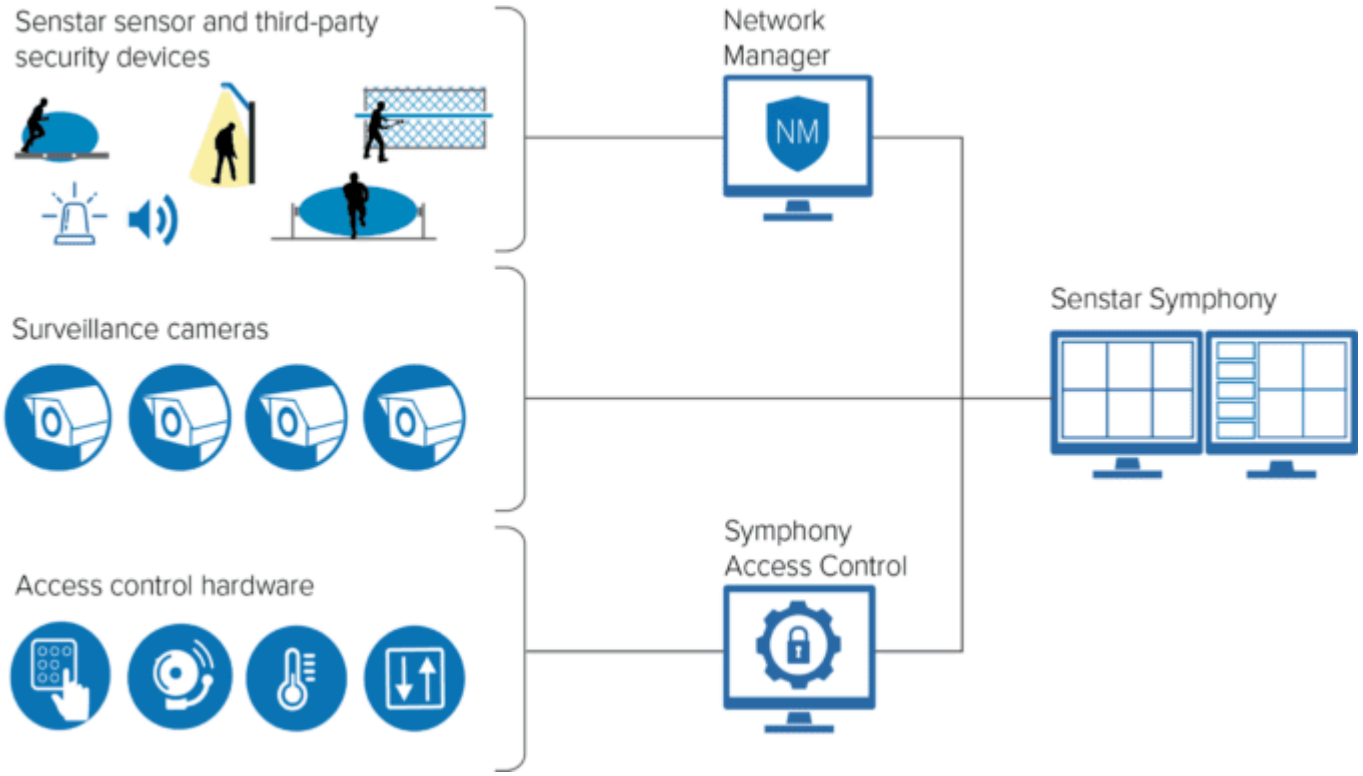
Senstar Integration and Alarm Configuration Guide

Introduction

This guide describes how to **integrate** and **configure** the Senstar Video Management System (VMS) within your security operations framework. It covers the key steps for configuring device connectivity, and integration with monitoring platforms. The goal is to enable seamless event telemetry from Senstar into GCX-ONE platform and ensure effective alerting, video correlation, and operational visibility.

System Overview

Senstar's video management offerings include the VMS software platform (e.g., Senstar Symphony Common Operating Platform) and supporting modules for cameras, video analytics, access control, and perimeter intrusion detection.



When used in your secure environment, the Senstar VMS acts as the **central video & event engine**:

It aggregates video streams from cameras around the site.

It collects alarms and status from sensors, analytics, and connected I/O.

It provides a unified UI for operators and integrates with external platforms.

System Architecture

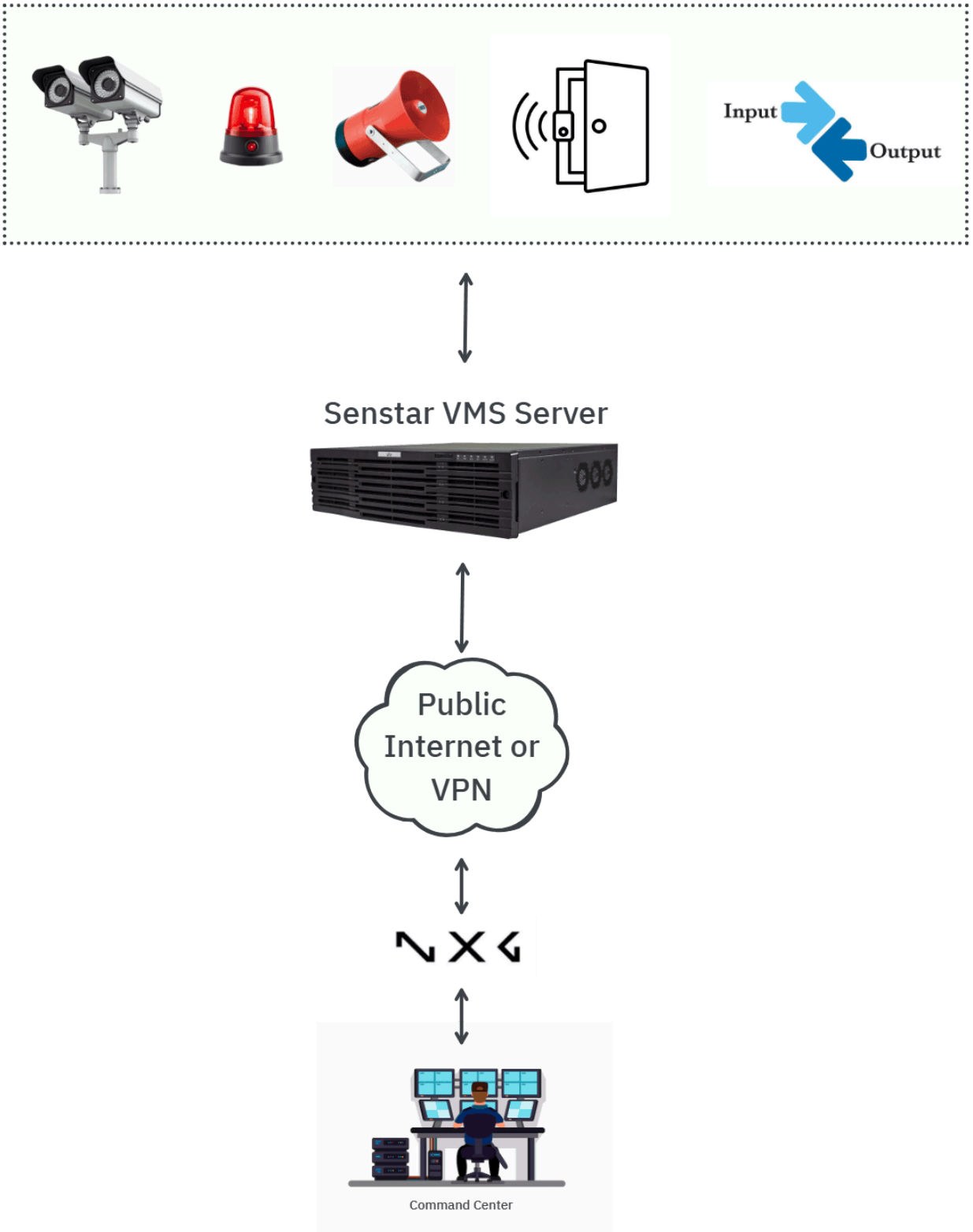
In a typical deployment:

Cameras (IP/analog) stream into the Senstar VMS server (on-prem / appliance).

Connected security sensors feed I/O and telemetry into the VMS or its sensor integration module.

The VMS generates events (e.g., intrusion detection, sensor alarm, video analytic alert) that are forwarded to the central monitoring system via API, TCP, or SDK.

GCX-ONE receives those events, perform false alarm filters, and triggers notifications to CMS accordingly.



Configuration Overview

Pre-Integration Requirements

Confirm the Senstar VMS version (8.10) and any required modules (e.g., video management, analytics).

Ensure network connectivity (either public network access or through VPN) between the VMS server and the monitoring platform (GCX-ONE).

Make sure the Mobile Bridge API Service is up and running. Detailed Senstar system configuration guide available here.

Create or identify an **integration user account** in Senstar with sufficient administrator privileges to access event streams and alarms.

Device Configuration in GCX-ONE

In the **GCX-ONE platform**, you can register the **Senstar VMS instance** as a device of type **“SenStar”**.This configuration establishes a secure communication channel between GCX-ONE and the Senstar Video Management System, enabling event synchronization, video stream access, and command & control.

Key Configuration Parameters

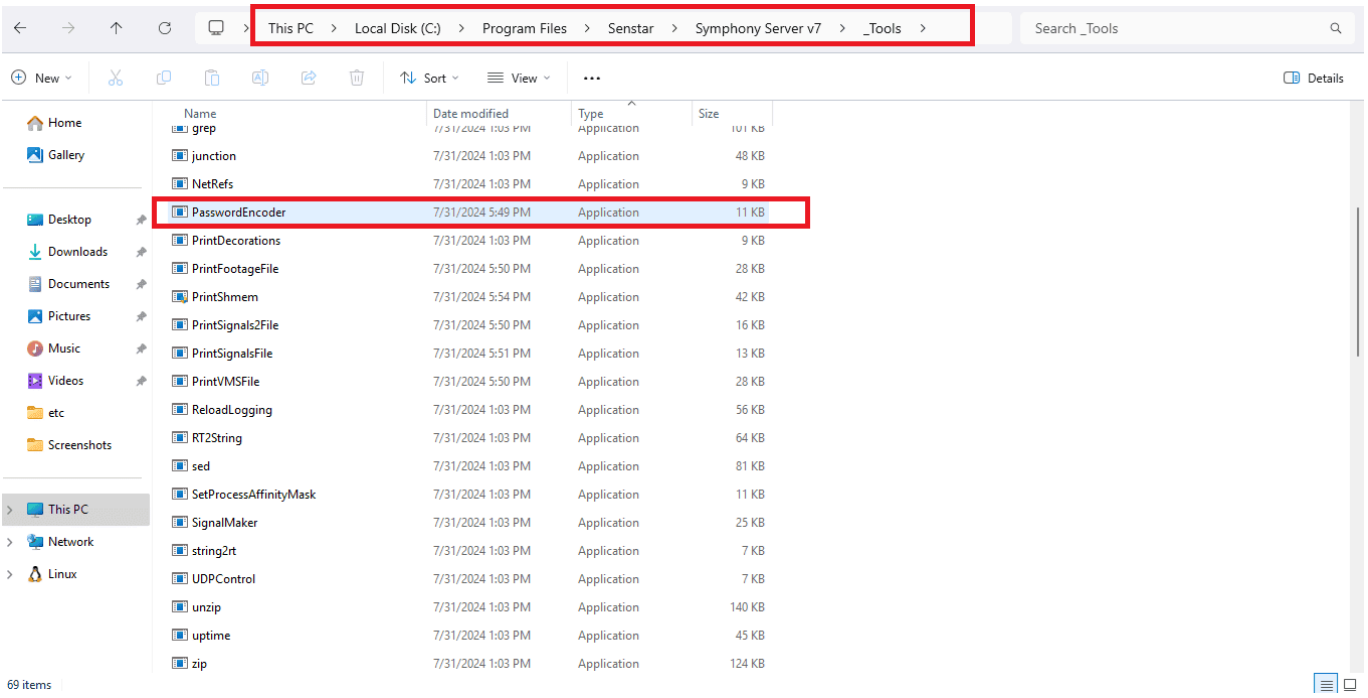
Streaming Key Generation

The **Senstar VMS integration** requires an **encoded streaming password** to securely access RTSP-based live and recorded video streams. This mechanism enhances security by avoiding direct exposure of the plain-text password during stream authentication.

Steps to Generate the Encoded Password

Locate the Password Encoder Utility The password encoding tool, PasswordEncoder.exe, is installed as part of the Senstar Symphony Server package.It cannot be downloaded separately. Ensure that the **Senstar Symphony Server** is installed on your system before proceeding.

Path:

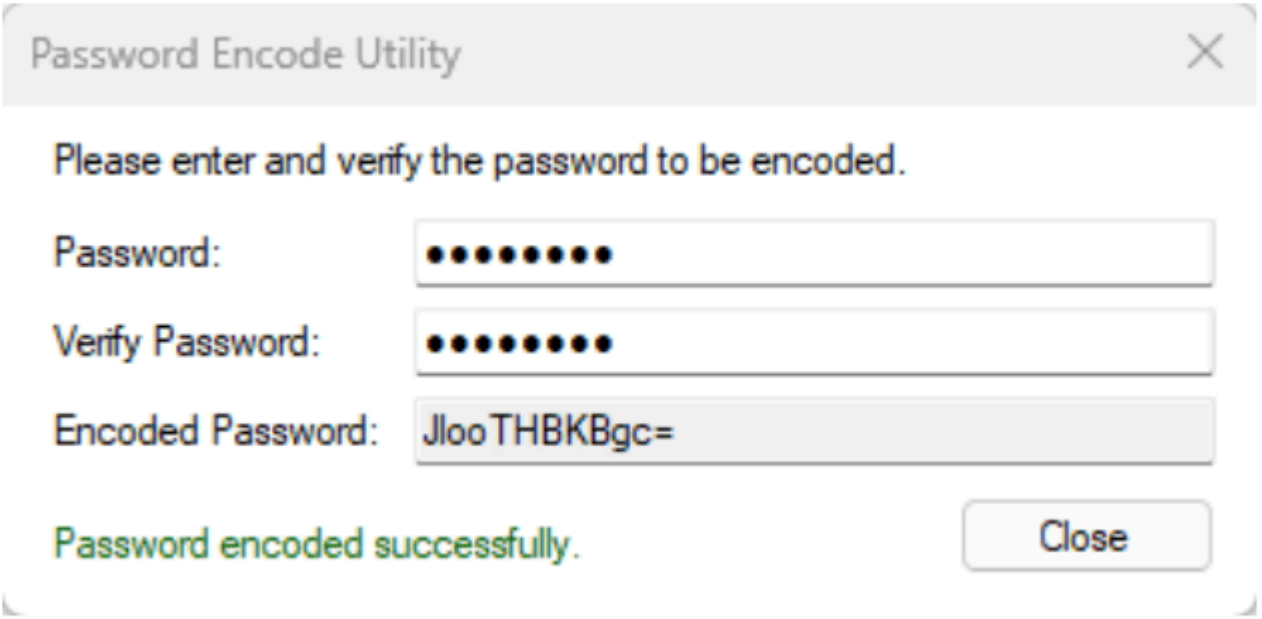


C:\Program Files\Senstar\Symphony Server V7_Tools\PasswordEncoder.exe

Double-click **PasswordEncoder.exe** to launch the encoder utility.

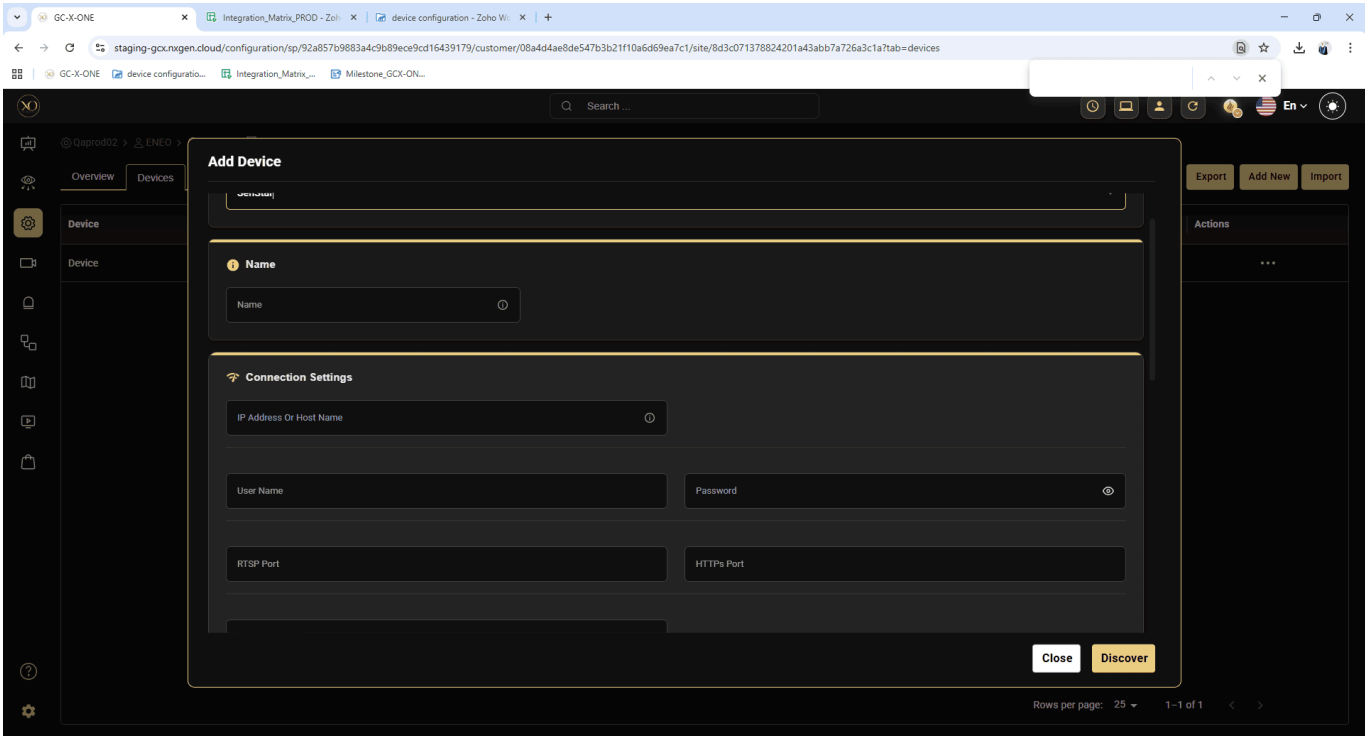
Enter the Device Password In the utility window, input the original (plain-text) device password that is configured in the Senstar VMS.

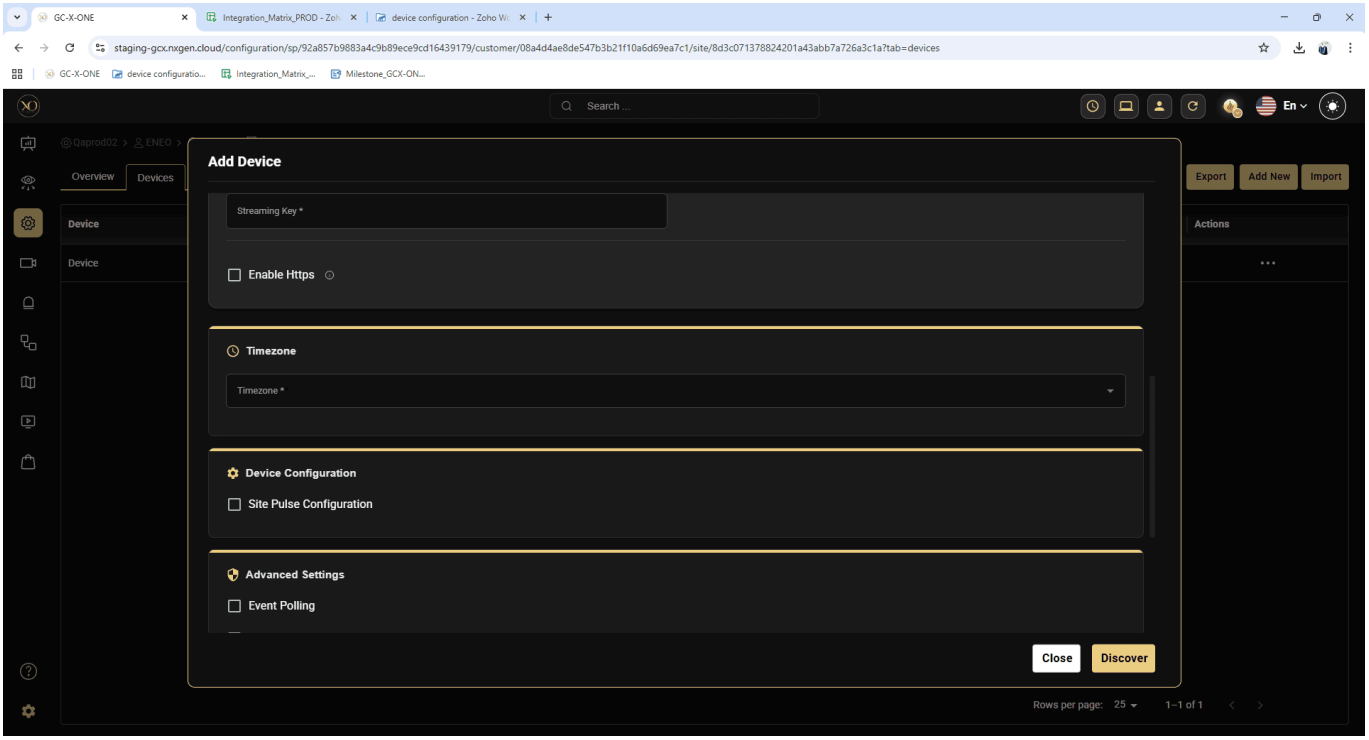
Copy the Encoded Password Once the password is encoded, copy the resulting **Encoded Password** string.



Paste the encoded password value into the **Streaming Key** field within the **GCX-ONE Device Configuration** page for your Senstar VMS instance.

Configuration Screenshot sample





Events and Rules Configuration

Configuring Senstar for Analytics and System Alarms

The **Senstar VMS** platform offers a comprehensive suite of **video analytics** that deliver real-time alerts that can be integrated with GCX-ONE for unified monitoring.

These analytics include features such as motion detection, camera tampering, license plate recognition, object tracking (people and vehicles), indoor/outdoor people counting, and advanced sensor fusion.

Before you can enable a video analytic for a camera, you must assign the appropriate **analytic license** to the device and configure any required parameters — such as drawing processing masks or calibrating the camera's view.For detailed instructions on how to configure these analytics and system alarms, please refer to the official **Senstar Configuration Guide** provided by Senstar: Configuration Guide

Senstar Supported Analytics feature

Action Set Configuration Guide

The Action Set configuration in **Senstar Symphony VMS** defines what actions the system performs when specific rules or events are triggered — for example, sending a notification, activating an external device, or forwarding an alarm to a third-party platform.

In the **GCX-ONE platform integration**, Action Sets play a key role in ensuring that all analytic or system events from Senstar are forwarded reliably to GCX-ONE over a **TCP connection**. Once Events and Rules are configured within Senstar, an Action Set must be created and linked to those rules to complete the integration.

Pre-requisites

Before configuring Action Sets for GCX-ONE:

In GCX-ONE: navigate to **Device** → **Edit** → **Unique Reference Number**.

The **Senstar VMS device** must already be added to GCX-ONE and have a valid **Unique Reference Number (deviceId)**.

Events and Rules should be configured in Senstar for the cameras or analytics you want to monitor.

Network access must be allowed from the Senstar VMS to the GCX-ONE cloud server on the specified TCP port.

Creating an Action Set in Senstar

Follow the standard Senstar procedure to create a new Action Set:

Open Senstar Symphony Server Configuration Client.

Go to **Rules and Actions** → **Action Sets**.

Click **New Action Set**.

Enter a descriptive name for the Action Set (e.g., GCX-ONE_Alarm_Forwarder).

Under **Actions**, click **Add Action** → select **TCP**.

GCX-ONE-Specific TCP Configuration

In the TCP Message configuration, apply the following settings:

Message JSON Format

{

"deviceId": "<>",

"Camera": "%c",

"cameraName": "%e",

"file": "%f",

"RuleName": "%b",

"Policy": "%o",

"metadata": "%md",

"AlarmTime": {

```
"year": "%YYYY",

"month": "%MM",

"date": "%DD",

"hour": "%hh",

"minute": "%mm",

"second": "%ss",

"ms": "%fff"
```

}

}

Parameter Description

Placeholder Reference (Senstar Syntax)

The placeholders such as %c, %e, %f, etc. are dynamic variables replaced automatically by Senstar when an event occurs:

Testing and Verification

Trigger a known rule in Senstar (e.g., motion or analytics alert).

In GCX-ONE, verify that the corresponding event appears in Video activity search or device level alarm dashboard.

If no events appear, confirm network reachability to 18.158.38.178:19232 and verify that the Action Set is active.

Best Practices

Always test connectivity with telnet 18.158.38.178 19232 before saving.

Ensure firewall or outbound network policies allow TCP traffic to the GCX-ONE endpoint.

Avoid modifying placeholder syntax — any incorrect character will break JSON parsing.

Use UTC timestamps where possible to maintain time sync between Senstar and GCX-ONE.

Integrations Features Overview

The GCX-ONE platform integrates seamlessly with the **Senstar Symphony VMS** through a combination of **Mobile Bridge APIs**, **RTSP streaming**, and **TCP-based alarm forwarding**. This integration enables unified management, monitoring, and analytics for all connected cameras and sensors within the Senstar ecosystem.

Summary

By following this guide, you will be able to configure your Senstar VMS instance to feed live video and alarm event data into your GCX-ONE monitoring ecosystem. This integration helps unify video surveillance, sensor alarms, and makes your monitoring operations more responsive and holistic.